

NOTEBOOK

OVERVIEW

ZJU-CHINA2026





Notebook — Mitochondrial Therapy Module

Timeline of the Work

Date (2026)	Work performed	Record
6 July	Transformation of pET28a-MMP13sp and pET28a-Control into E. coli BL21(DE3); selection on kanamycin plates	1
7–11 July	Small-scale expression of the responsive and control fusion proteins; IPTG gradient screening with Western-blot-assisted detection	2
12–15 July	Ni-NTA affinity purification of the two fusion proteins; SDS-PAGE analysis of the purification fractions	3
16–19 July	Dialysis of the purified proteins and determination of the protein concentrations	4
20–21 July	TEV digestion of the SUMO tag and recovery of the functional peptide	5
22–24 July	Verification of the DogTag ligation activity of the recovered peptide; lyophilisation for concentration	6
25 July–1 August	Recovery, culture and maintenance of the recipient cell line (practice; no experimental data collected)	7
2–3 August	Isolation of mitochondria; Western-blot confirmation of COX IV and TOMM20	8
4–5 August	Preparation of the maleimide-functionalised liposomes; assembly of the liposome–mitochondria complexes	9
6 August	Confocal fluorescence imaging of the complexes (MitoTracker Green / DiD)	10
7–15 August	TEM characterisation of the liposomes, mitochondria and complexes	11
16–18 August	Preliminary evaluation of the Cys–maleimide coupling of the functional peptide to the liposomes	12
19 August	SnoopLigase-mediated ligation of MMP13-DogTag and Control-DogTag to GFP-SnoopTag	13
20–22 August	MMP-13 digestion time course of the responsive and control reporter substrates	14
23–25 August	Cell-delivery evaluation of the liposome–mitochondria complexes by fluorescence imaging	15
26–30 August	CCK-8 cell-viability assay of the functional peptide (5 and 10 μM; 24, 48 and 96 h)	16

Experiment Records

1 Transformation of the pET28a-MMP13sp and pET28a-Control plasmids into E. coli BL21(DE3)

Date: 6 July 2026

Procedures

1. The two expression plasmids of this module, pET28a-MMP13sp (responsive construct) and pET28a-Control (scrambled control), are both 5691 bp and share the same backbone, T7 promoter and kanamycin-resistance gene. Each encodes a fusion protein consisting, from N-terminus to C-terminus, of a 6×His tag, a SUMO tag, a TEV protease cleavage site, a cysteine coupling site, the functional peptide and a DogTag, with GGGGS linkers flanking the peptide segment.
2. The responsive construct encodes the MMP-13-cleavable peptide RRGPLGLSLGKR; the control construct carries the scrambled sequence RKGLSLGLPGRR. Both peptides are 12 amino acids long and identical in composition, differing only in the order of the residues, so that any difference in protease sensitivity can be attributed to the sequence arrangement.
3. Chemically competent BL21(DE3) cells were thawed on ice for 5–10 min; 50 µL aliquots were mixed with 1–2 µL of plasmid DNA. A negative control without DNA was set up in parallel.
4. The mixtures were kept on ice for 30 min, heat-shocked at 42 °C for 45–90 s and returned to ice for 2 min.
5. 800 µL–1 mL of pre-warmed LB broth without antibiotic was added, and the cells were recovered at 37 °C, 200 rpm for 45–60 min.
6. 50–200 µL of each recovery culture was spread onto LB agar containing kanamycin; the plates were incubated inverted at 37 °C for 12–16 h.

Note

Both plasmids gave well-separated single colonies on the kanamycin plates, whereas the negative-control plate remained clean, confirming that the transformations were successful and that the selection was effective. Single colonies were picked and used to inoculate liquid cultures for the expression trials (record 2).

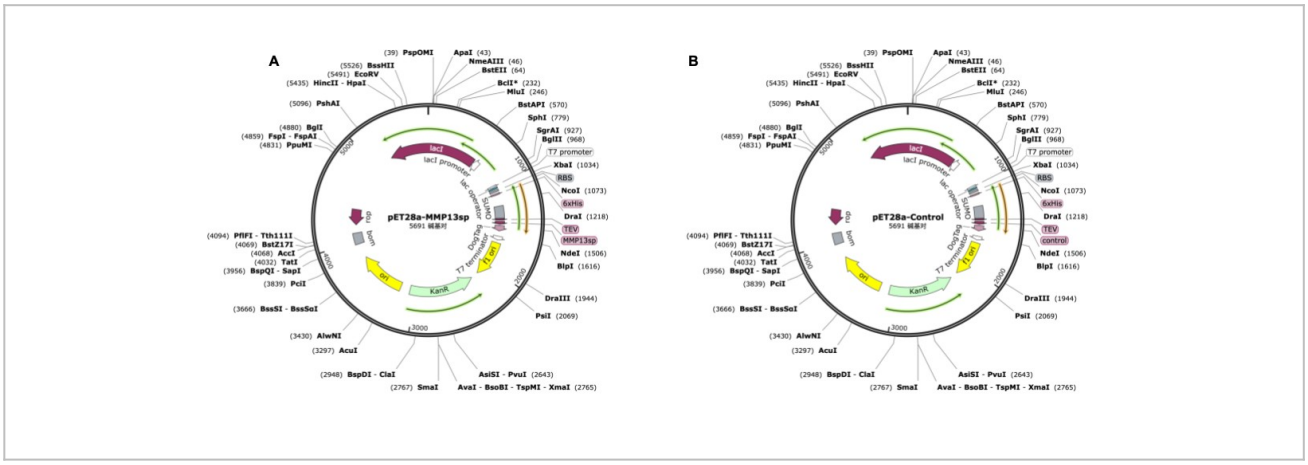


Figure 1. Maps of the two expression plasmids used in this module. (A) pET28a-MMP13sp; (B) pET28a-Control. Both plasmids are 5691 bp and carry the T7 promoter, kanamycin resistance and the N-terminal 6×His–SUMO–TEV–Cys–peptide–DogTag expression

cassette; the responsive construct encodes the MMP-13-cleavable peptide RRGPLGLSLGKR and the control construct the scrambled sequence RKGLSLGLPGRR.

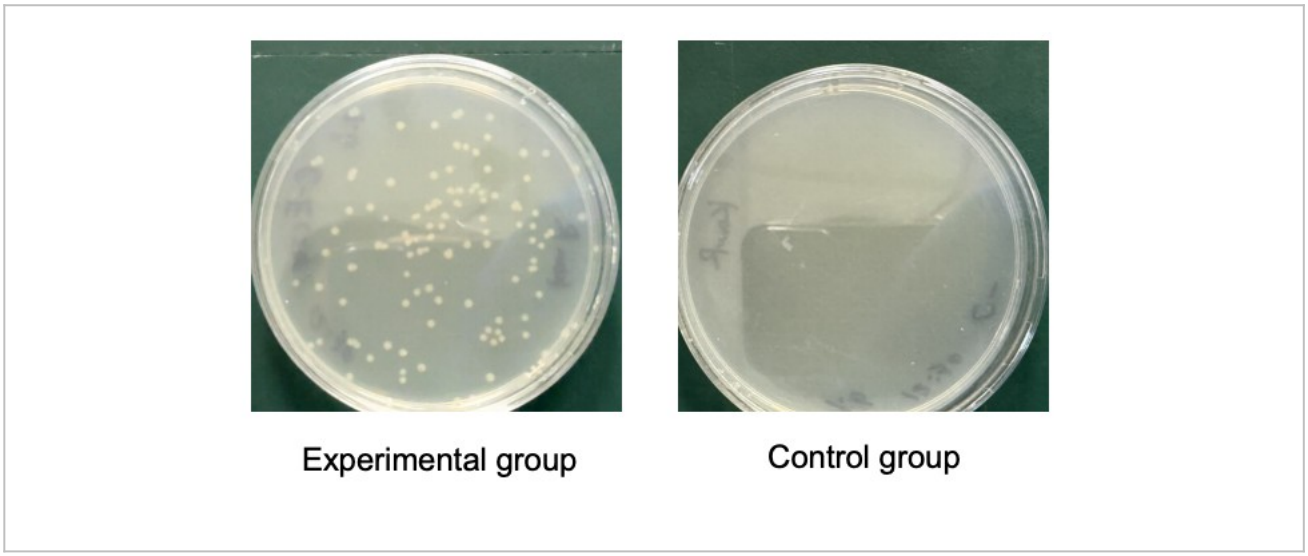


Figure 2. Selection of transformants on LB–kanamycin agar. Well-separated colonies appeared on the plasmid-transformation plate, whereas the negative-control plate without plasmid remained clean, confirming the success of the transformations and the effectiveness of the kanamycin selection.

2 Small-scale expression of the responsive and control fusion proteins and IPTG gradient screening

Date: 7–11 July 2026

Procedures

1. Single colonies of each transformant were inoculated into 5 mL LB–kanamycin starter cultures and grown at 37 °C, 200 rpm overnight.
2. The starter cultures were diluted 1:100 into fresh LB–kanamycin medium and grown to mid-exponential phase.
3. Expression of the two fusion proteins was induced with IPTG at a range of concentrations, and samples were taken before and after induction for whole-cell SDS-PAGE to compare the expression levels.
4. Because the target bands in the early gels were weak and poorly resolved, Western blot was used to assist the detection of the target proteins, and the expression and electrophoresis conditions were adjusted accordingly.

Note

In the gradient screening the target bands were most prominent at 0.25 and 0.5 mM IPTG, and these two conditions were selected for the scale-up expression. Western-blot-assisted detection confirmed the expression of both fusion proteins under the adjusted conditions.

3 Ni-NTA affinity purification of the two fusion proteins

Date: 12–15 July 2026

Procedures

1. The responsive and control transformants were grown in scale-up cultures and induced with IPTG at the two selected concentrations (0.25 and 0.5 mM); the cells were harvested by centrifugation.

2. The cell pellets were resuspended in lysis buffer and disrupted on ice; the lysates were clarified by centrifugation, and samples of the clarified lysates were retained for analysis.
3. Each lysate was purified by Ni-NTA affinity chromatography; the flow-through, wash and eluate fractions were collected during purification.
4. The lysate, flow-through, wash and eluate fractions of both purifications were analysed by SDS-PAGE with Coomassie staining.

Note

For the samples induced with 0.25 mM IPTG, enriched bands were observed in the eluate fractions of both purifications at 15–25 kDa, consistent with the expected molecular-weight range of the target fusion proteins, supporting the expression and enrichment of both the responsive and the control protein.

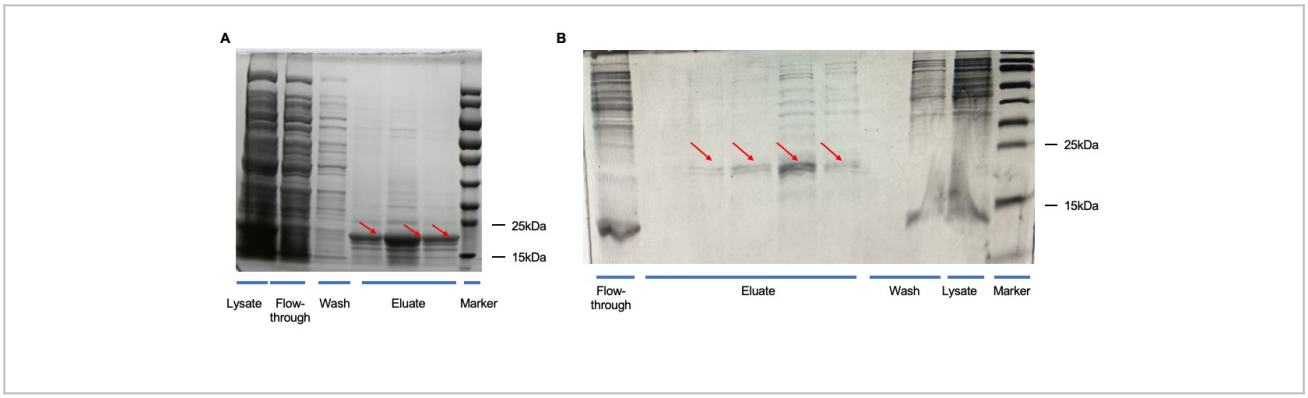


Figure 3. Ni-NTA purification of the two fusion proteins (samples induced with 0.25 mM IPTG). SDS-PAGE of the lysate, flow-through, wash and eluate fractions of (A) the responsive and (B) the control construct; enriched bands at 15–25 kDa correspond to the target fusion proteins.

4 Dialysis and concentration determination of the purified proteins

Date: 16–19 July 2026

Procedures

1. The eluate fractions containing the target proteins were pooled and dialysed at 4 °C to remove imidazole and salts; the dialysed samples were stored for the subsequent experiments.
2. The protein concentrations of the dialysed samples were determined by the BCA method.

Note

One batch of the responsive fusion protein was measured at 0.43 mg/mL and the control protein at 0.52 mg/mL, providing the experimental material for the tag removal, the protein ligation and the MMP-13 digestion assays.

5 TEV digestion of the SUMO tag and recovery of the functional peptide

Date: 20–21 July 2026

Procedures

1. To remove the SUMO tag from the responsive fusion protein and release the responsive peptide for the later ligation and functional assays, the purified fusion protein was incubated with TEV protease at 30 °C.

2. Samples were taken after 1, 2 and 4 h of digestion; a reaction without TEV protease was set up in parallel as the undigested control.
3. All samples were analysed by SDS-PAGE.

Note

At all three digestion time points a band of the released SUMO tag appeared in the 15–25 kDa region (red arrows), migrating differently from the undigested fusion protein, supporting TEV-mediated removal of the SUMO tag. The bands indicated by yellow arrows correspond to the added TEV protease (~27 kDa). These results justified the subsequent separation of the responsive peptide and the protein-ligation experiments.

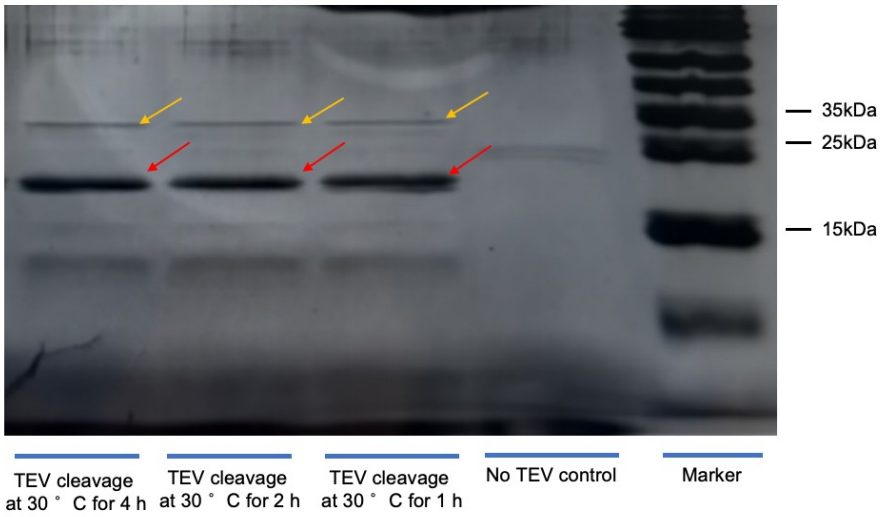


Figure 4. TEV digestion of the purified responsive fusion protein at 30 °C. Lanes: no-TEV control and digestion for 1, 2 and 4 h. Red arrows, released SUMO tag (15–25 kDa); yellow arrows, added TEV protease (~27 kDa).

6 Verification of the ligation activity of the recovered peptide and concentration by lyophilisation

Date: 22–24 July 2026

Procedures

1. To verify the ligation capability of the recovered functional peptide, the DogTag-containing peptide sample was mixed with the GFP-SnoopTag reporter protein and SnoopLigase and incubated at room temperature for 1 h.
2. The enzymatic ligation of SUMO-DogTag with GFP-SnoopTag was run in parallel as the positive control, and a reaction without SnoopLigase served as the negative control.
3. The reaction mixtures were analysed by SDS-PAGE.
4. Because the concentration of the recovered peptide was low, the remaining sample was lyophilised to concentrate it for the later ligation and digestion assays.

Note

Compared with the negative control, a new higher-molecular-weight band appeared above the GFP-SnoopTag main band in the peptide lane (red arrow), consistent with the formation of a peptide–GFP-SnoopTag ligation product.

The band at 10–15 kDa (yellow arrow) corresponds to SnoopLigase. The recovered sample therefore contains a functional peptide with DogTag-mediated ligation activity, providing a gel-detectable reporter substrate for the MMP-13 responsiveness study.

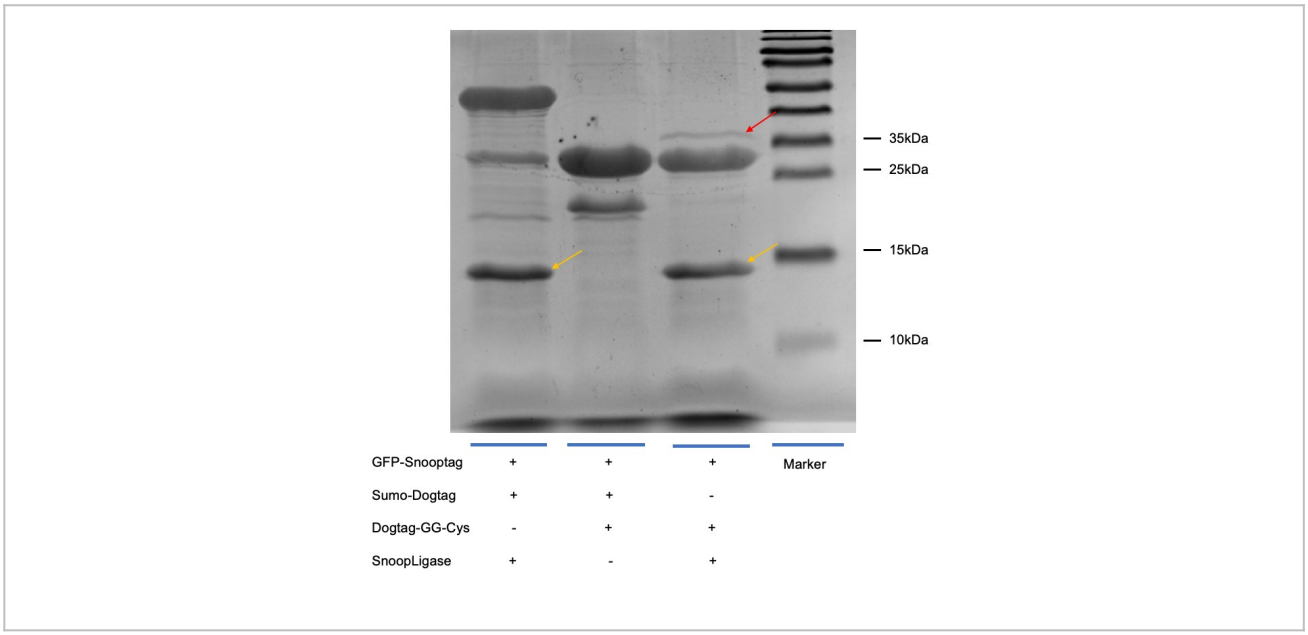


Figure 5. DogTag ligation activity of the recovered functional peptide. The peptide was reacted with GFP-SnoopTag and SnoopLigase for 1 h at room temperature; the SUMO-DogTag reaction served as the positive control and the reaction without SnoopLigase as the negative control. Red arrow, new band above the GFP-SnoopTag main band corresponding to the peptide–GFP-SnoopTag ligation product; yellow arrow, SnoopLigase (10–15 kDa).

7 Recovery and culture of the recipient cell line

Date: 25 July–1 August 2026

Procedures

1. The recipient cell line used for the delivery and viability assays (____ cells) was recovered from cryostock and expanded in complete culture medium.
2. The cells were passaged routinely; morphology, confluence and absence of contamination were checked by microscopy at every passage.
3. Handling, counting and seeding for fluorescence-imaging and 96-well-plate experiments were practised in preparation for the delivery and CCK-8 assays.

Note

This was a training and maintenance period for the cell work of this module; no experimental data were collected. The culture remained healthy and was carried forward by routine passaging to the experiments scheduled for late August.

8 Isolation of mitochondria and confirmation by Western blot

Date: 2–3 August 2026

Procedures

1. Mitochondria were isolated from ____ cells by differential centrifugation: the sample was homogenised in ice-cold isolation buffer, nuclei and debris were removed by low-speed centrifugation, and mitochondria were pelleted at high speed and washed.
2. The input, supernatant and precipitate fractions of the isolation were collected for analysis.
3. The fractions were analysed by Western blot for the inner-membrane marker COX IV and the outer-membrane marker TOMM20.

Note

Signals for both COX IV and TOMM20 were detected in the precipitate fraction, supporting the presence of mitochondrial components in the isolated sample and providing the material basis for the assembly of the liposome–mitochondria complexes.

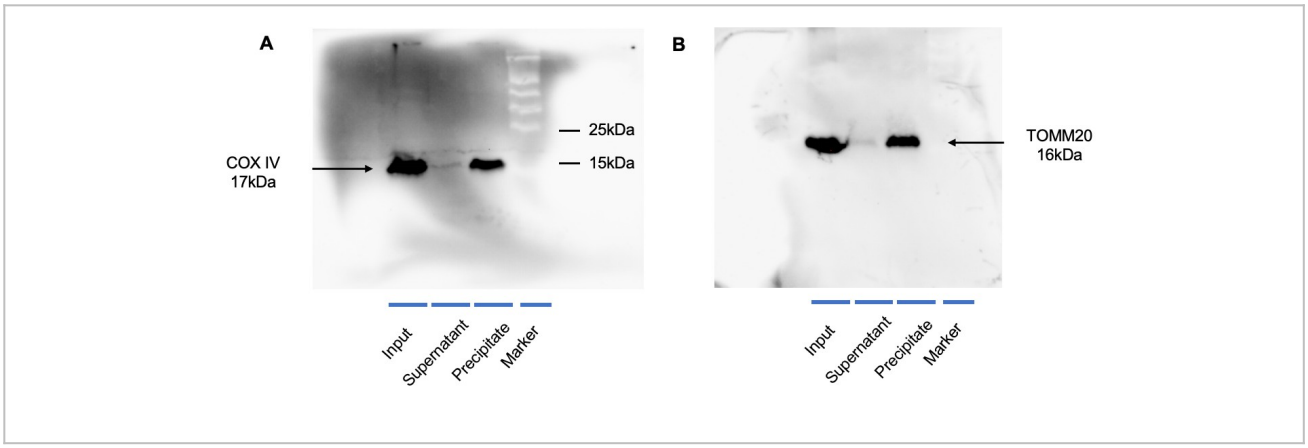


Figure 6. Western-blot confirmation of the mitochondrial isolation. (A) COX IV (inner-membrane marker, 17 kDa); (B) TOMM20 (outer-membrane marker, 16 kDa); lanes: input, supernatant and precipitate fractions. Signals in the precipitate fraction support the presence of mitochondrial components in the isolated material.

9 Preparation of the functionalised liposomes and assembly of the liposome–mitochondria complexes

Date: 4–5 August 2026

Procedures

1. Functionalised liposomes were prepared from DOPE, DOTAP and DSPE-PEG-Mal: the lipids were mixed at the designed ratio, dried and hydrated with buffer to form the vesicle suspension.
2. DSPE-PEG-Mal was included to display maleimide groups on the liposome surface for the later coupling of the cysteine thiol of the engineered peptide.
3. The appearance of the prepared suspension was documented; the liposomes were then co-incubated with the freshly isolated mitochondria to assemble the liposome–mitochondria complexes.

Note

The prepared liposome suspension had a milky-white appearance. The complexes assembled by co-incubation were taken forward to the confocal and TEM characterisation (records 10 and 11).

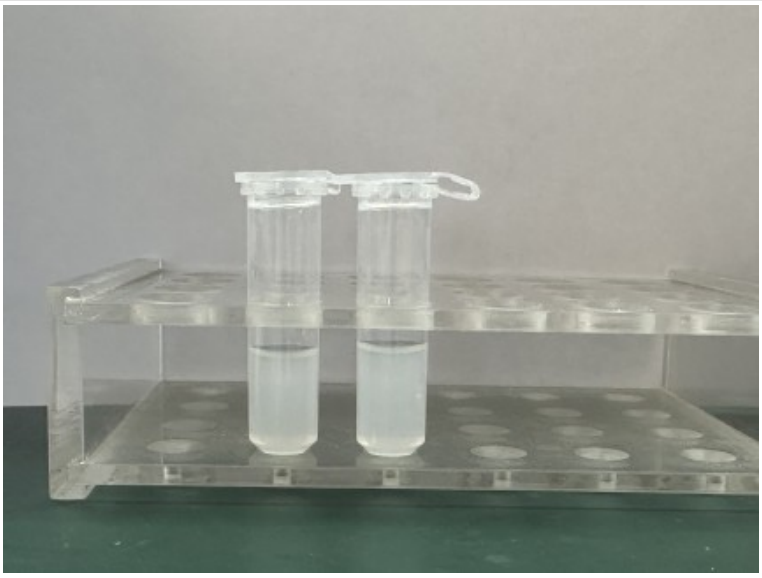


Figure 7. Appearance of the functionalised liposome suspension (milky white).

10 Confocal fluorescence characterisation of the liposome–mitochondria complexes

Date: 6 August 2026

Procedures

1. The liposomes were labelled with DiD, and the freshly isolated mitochondria were labelled with MitoTracker Green.
2. The labelled components were co-incubated to assemble the complexes, which were imaged by confocal fluorescence microscopy; the green channel, the red channel and the bright-field image were acquired and merged.

Note

The mitochondria labelled with MitoTracker Green showed green fluorescence and the DiD-labelled liposomes red fluorescence, and the two signals overlapped at multiple positions, giving yellow–orange regions in the merged image. This supports a spatial association between the liposomes and the mitochondria; whether an encapsulating structure had formed was further evaluated by TEM.

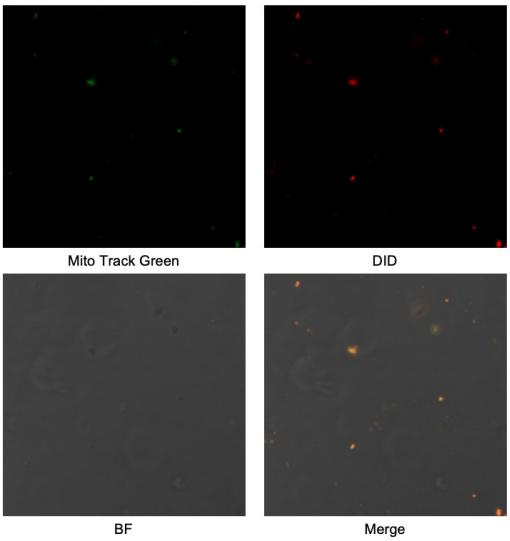


Figure 8. Confocal fluorescence imaging of the liposome–mitochondria complexes. Green, MitoTracker Green-labelled mitochondria; red, DiD-labelled liposomes; BF, bright field; merge. Overlapping green and red signals give yellow–orange regions, supporting a spatial association between the liposomes and the mitochondria.

11 TEM characterisation of the liposome–mitochondria complexes

Date: 7–15 August 2026

Procedures

1. The functionalised liposomes, the free mitochondria and the assembled liposome–mitochondria complexes were prepared on electron-microscopy grids, negatively stained and imaged by TEM.
2. The morphology and structure of the three samples were compared.

Note

Vesicle-like structures and membrane outlines were observed in the liposome sample. In the complexes, a peripheral membrane-like structure surrounding internal components was observed, consistent with the expected configuration of liposomes encapsulating mitochondria. Together with the overlapping fluorescence signals in record 10, these results support the successful construction of the liposome–mitochondria complexes.

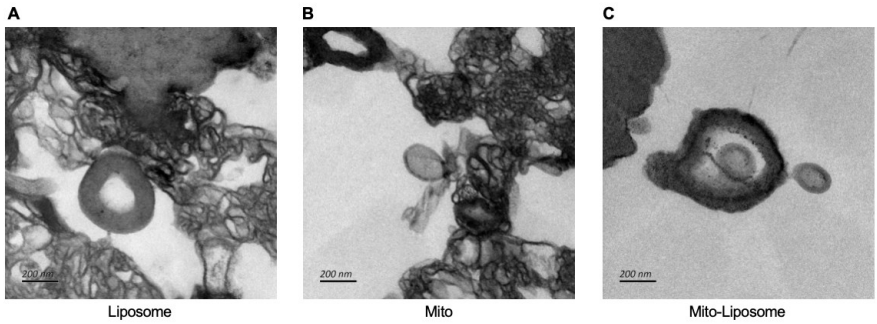


Figure 9. TEM micrographs of (A) functionalised liposomes, (B) free mitochondria and (C) liposome–mitochondria complexes. A peripheral membrane-like structure surrounding internal components is visible in (C), consistent with the configuration of liposomes encapsulating mitochondria (scale bar: 200 nm).

12 Preliminary evaluation of the coupling of the functional peptide to the liposomes

Date: 16–18 August 2026

Procedures

1. To connect the liposome vehicle with the protein-ligation system, the cysteine thiol of the concentrated functional peptide was reacted with the maleimide groups displayed on the liposome surface.
2. Ultrafiltration was used after the reaction to separate the unbound free peptide from the liposome fraction.
3. Protein quantification of the fractions before and after the separation was attempted in order to evaluate the extent of coupling.

Note

The ultrafiltration step suffered from unstable recovery and sample loss, and the liposome suspension interfered with the colorimetric protein assay, so the measured decrease of free peptide could not be attributed unambiguously to covalent coupling, non-specific adsorption or separation losses. A reliable coupling efficiency therefore could not be calculated at this stage. Future work will optimise the separation and quantification methods, include recovery controls and liposomes without maleimide groups as negative controls, and establish a robust workflow for verifying the coupling, providing the basis for integrating the delivery vehicle with the SELP scaffold.

13 SnoopLigase-mediated ligation activity of MMP13-DogTag and Control-DogTag

Date: 19 August 2026

Procedures

1. To verify the ligation activity of the DogTag module in the two constructs, the responsive protein (MMP13-DogTag) and the scrambled control protein (Control-DogTag) were each mixed with the GFP-SnoopTag reporter protein.
2. For each pairing, reactions with and without SnoopLigase were set up in parallel.
3. The reaction mixtures were incubated and analysed by SDS-PAGE for the formation of ligation products.

Note

A higher-molecular-weight band at approximately 55 kDa appeared in both the responsive and the control reactions containing SnoopLigase (red arrows), whereas no corresponding band was observed in the reactions without the enzyme. The mobility shift is consistent with the increase in molecular weight expected from the ligation of each protein to GFP-SnoopTag, supporting SnoopLigase-mediated formation of the ligation products. Both the responsive and the scrambled control construct therefore retain DogTag-mediated ligation activity, providing the experimental basis for comparing the MMP-13 digestion of the two reporter substrates.

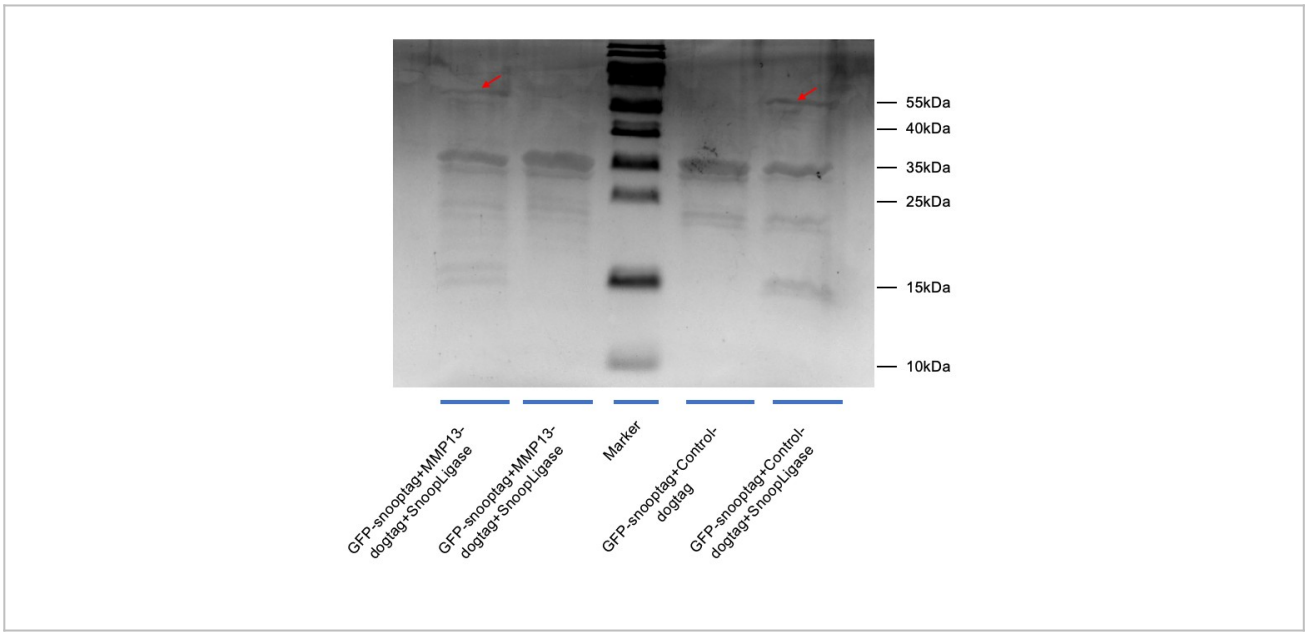


Figure 10. SnoopLigase-mediated ligation of MMP13-DogTag and Control-DogTag to GFP-SnoopTag. Red arrows, higher-molecular-weight bands (~55 kDa) in the reactions containing SnoopLigase; no corresponding bands appear in the reactions without the enzyme.

14 MMP-13 digestion time course of the responsive and control reporter substrates

Date: 20–22 August 2026

Procedures

1. The responsive peptide (Pep) and the scrambled control peptide (Control) were each ligated to GFP-SnoopTag to build reporter substrates detectable by gel electrophoresis.
2. The two substrates were digested with MMP-13, and samples were taken at 0, 2, 4, 6 and 24 h.
3. The time-course samples were analysed by SDS-PAGE, and the band patterns of the two groups were compared.

Note

The Control group retained a slower-migrating band at approximately 35 kDa corresponding to the undigested GFP–peptide ligation product (red arrows), whereas the corresponding band in the Pep group was weak or not clearly detected. This band pattern is consistent with the cleavage of the responsive ligation product and indicates that the responsive peptide is cleavable by MMP-13. Because a difference between the two groups was already present in the 0 h samples, pre-digestion substrate controls and no-enzyme controls are still required to confirm that the difference arises from MMP-13 cleavage rather than from different initial compositions of the two ligation products.

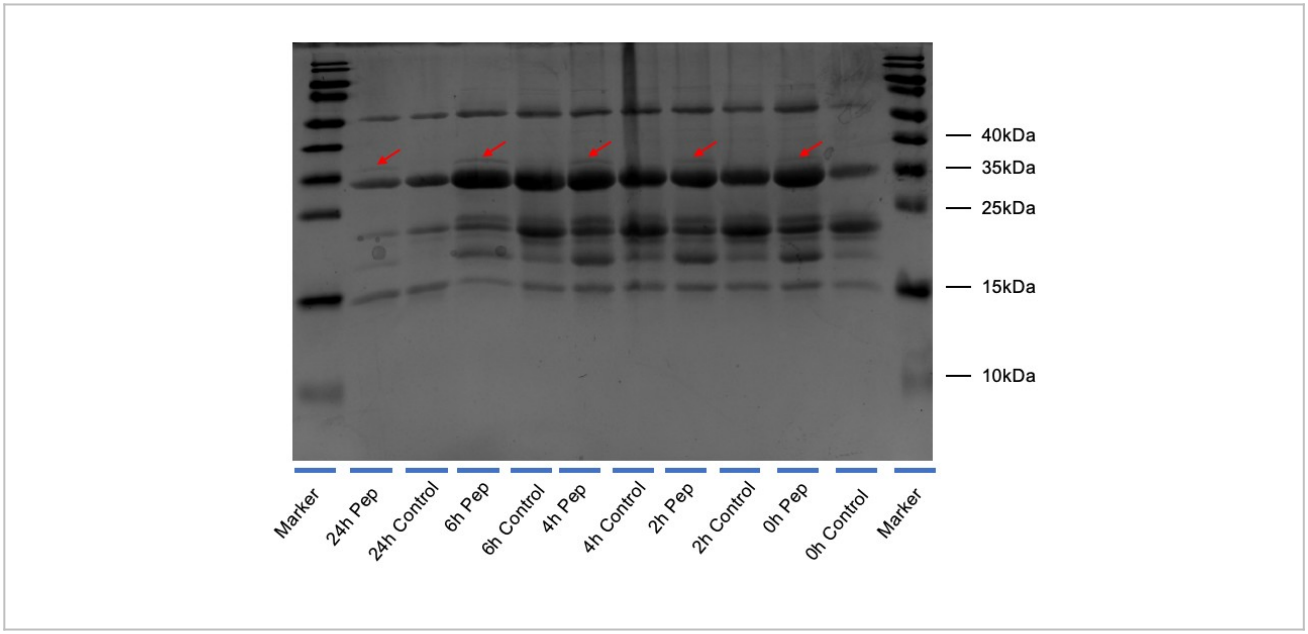


Figure 11. MMP-13 digestion time course of the reporter substrates. Samples were taken at 0, 2, 4, 6 and 24 h for the responsive (Pep) and scrambled (Control) substrates. Red arrows, the ~35 kDa band of the undigested GFP–peptide ligation product, retained in the Control group but weak or absent in the Pep group.

15 Cell delivery of the liposome–mitochondria complexes

Date: 23–25 August 2026

Procedures

1. The freshly isolated mitochondria were labelled with MitoTracker Green.
2. The labelled mitochondria were co-incubated with the functionalised liposomes to obtain the liposome–mitochondria complexes.
3. The complexes were added to the recipient-cell culture, and a control group without the complexes was set up in parallel.
4. The cells were stained with Hoechst 33342 and imaged by bright-field and fluorescence microscopy to locate the labelled mitochondria.

Note

Green fluorescence was observed within the cell boundary and in the perinuclear region of the complex-treated group (panel A), whereas no corresponding green signal was detected in the control group (panel B). These results indicate that the exogenous mitochondria of the complexes can enter the recipient cells, providing preliminary imaging support for the cell-delivery capability of the liposome–mitochondria complexes.

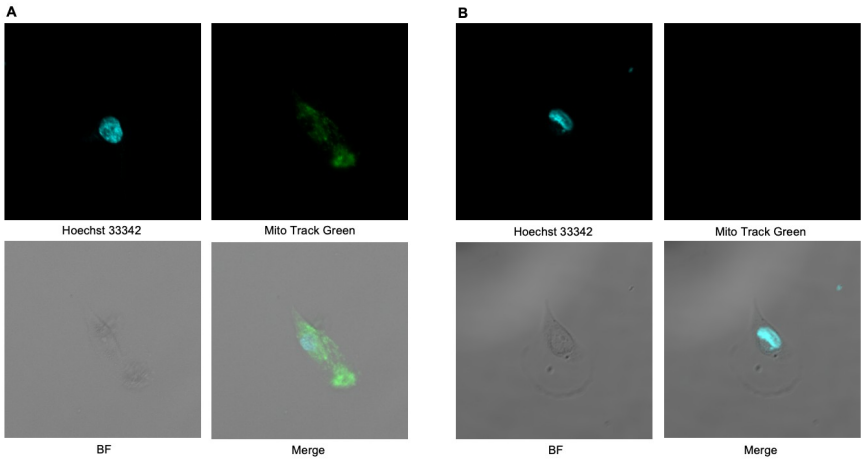


Figure 12. Cell delivery of the liposome–mitochondria complexes. Blue, Hoechst 33342 (nuclei); green, MitoTracker Green (exogenous mitochondria); BF, bright field; merge. (A) Complex-treated cells show green fluorescence within the cell boundary and around the nuclei; (B) the control group shows no corresponding green signal.

16 CCK-8 biosafety assay of the functional peptide

Date: 26–30 August 2026

Procedures

1. Recipient cells were seeded in 96-well plates; an untreated control group, a blank group and two peptide-treatment groups (5 and 10 μM) were set up.
2. Cell viability was measured with the CCK-8 reagent 24, 48 and 96 h after the peptide addition; the absorbance at 450 nm was recorded and the relative cell viability was calculated.

Note

After 24 h, the relative viability of the 5 μM and 10 μM groups was approximately 70% and 55%, respectively; at 48 h, both groups reached approximately 85–90%; and at 96 h, both groups remained within approximately 85–95%. The lower viability at 24 h may be related to the state of the cells at the start of the experiment, and the present data cannot distinguish the influence of the initial cell state from that of the peptide. The assay provides preliminary in-vitro compatibility data for the functional peptide; repeat experiments with a controlled initial cell state are required to clarify the cause of the early viability change and to determine the appropriate dosing conditions.

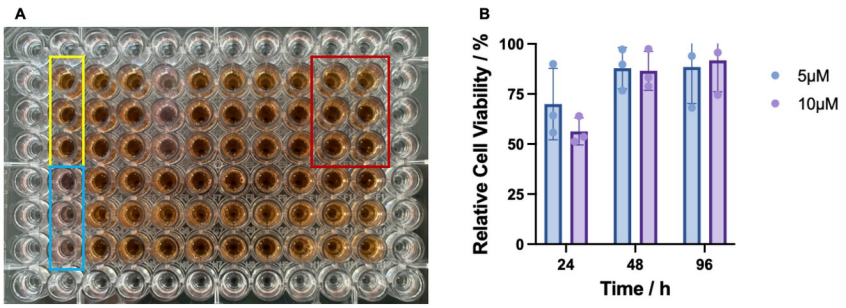


Figure 13. CCK-8 cell-viability assay of the functional peptide. (A) 96-well assay plate; (B) relative cell viability of the 5 μM and 10 μM treatment groups after 24, 48 and 96 h.

References

[1] Nakano T, Nakamura Y, Park J-H, et al. Mitochondrial surface coating with artificial lipid membrane improves the transfer efficacy. Communications Biology, 2022, 5: 745. doi:10.1038/s42003-022-03719-9. (lipid coating of isolated mitochondria with DOTAP/DOPE for improved transfer — methodology reference for the liposome–mitochondria complexes)

[2] Keeble AH, Wood DP, Howarth M. Design and Evolution of Enhanced Peptide–Peptide Ligation for Modular Transglutaminase Assembly. Bioconjugate Chemistry, 2023, 34(10): 1019–1036. doi:10.1021/acs.bioconjchem.3c00122. (SnoopTag2 / DogTag / SnoopLigase ligation system used for the peptide coupling and reporter assays)

[3] Tropea JE, Cherry S, Waugh DS. Expression and purification of soluble His6-tagged TEV protease. Methods in Molecular Biology, 2009, 498: 297–307. (TEV protease used for the removal of the SUMO tag)

[4] iGEM Team laboratory protocols P01–P14 (companion documents): plasmid transformation, IPTG-induced expression, SDS-PAGE, Western blot, Ni-NTA purification, dialysis, ultrafiltration, lyophilisation, ligation reaction and cell culture.